

Research Methodology in Public Administration

Proposal Writing and Report Writing

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Data Collection

Collecting Relevant Information to Answer Research Questions

Research Problem



Research Questions



Data Collection



Data Analysis



Findings

What is Data Collection?

Data collection is the systematic process of gathering relevant information to answer research questions and achieve research objectives.

The researcher must decide:

- ▶ What data are required?
- ▶ From whom will the data be collected?
- ▶ Where will the data be collected?
- ▶ How will the data be collected?
- ▶ When will the data be collected?

Data Collection in the Research Process



Major Sources of Data

Research data can broadly be divided into:

Primary Data

Data collected directly by the researcher for a specific research purpose.

Examples:

- ▶ Survey
- ▶ Interview
- ▶ Observation
- ▶ Focus Group Discussion

Secondary Data

Data that already exist and were collected by another individual or organization.

Examples:

- ▶ Census
- ▶ Government reports
- ▶ Research reports
- ▶ Administrative records

Quantitative and Qualitative Data

Quantitative Data

Numerical information that can be measured and analyzed statistically.

Examples:

- ▶ Age
- ▶ Income
- ▶ Number of employees
- ▶ Test scores
- ▶ Survey responses

Qualitative Data

Non-numerical information related to meanings, experiences, and perspectives.

Examples:

- ▶ Experiences
- ▶ Opinions
- ▶ Narratives
- ▶ Observations
- ▶ Documents

Major Data Collection Methods

Common methods include:

- ▶ Questionnaire
- ▶ Survey
- ▶ Interview
- ▶ Observation
- ▶ Focus Group Discussion
- ▶ Case Study
- ▶ Document Analysis
- ▶ Secondary Data Analysis

Important Principle

The data collection method should be appropriate for the research problem, objectives, and research design.

Questionnaire

A **questionnaire** is a structured set of questions designed to collect information from respondents.

Questionnaires may include:

- ▶ Closed-ended questions
- ▶ Open-ended questions
- ▶ Multiple-choice questions
- ▶ Rating scales
- ▶ Likert-scale questions

Types of Questionnaire Questions

Closed-Ended Question

How satisfied are you with the service?

Very Satisfied Satisfied Neutral Dissatisfied

Open-Ended Question

What challenges do you face while accessing public services?

Closed-ended questions are generally easier to quantify, whereas open-ended questions allow respondents to provide detailed answers.

Survey Research

A **survey** is a method of collecting standardized information from a sample of respondents.

Surveys can be conducted through:

- ▶ Face-to-face surveys
- ▶ Telephone surveys
- ▶ Online surveys
- ▶ Self-administered questionnaires

Example

A survey of citizens to measure their satisfaction with public service delivery.

Designing a Good Questionnaire

A good questionnaire should:

- ▶ Be related to the research objectives.
- ▶ Use clear and simple language.
- ▶ Avoid ambiguous questions.
- ▶ Avoid leading questions.
- ▶ Avoid unnecessary questions.
- ▶ Follow a logical sequence.
- ▶ Be tested before final data collection.

Interview

An **interview** is a method of collecting information through direct interaction between the researcher and participant.

Interviews are particularly useful for understanding:

- ▶ Experiences
- ▶ Perceptions
- ▶ Opinions
- ▶ Motivations
- ▶ Meanings

Types of Interviews

Structured

Predefined questions are asked in a fixed order.

Semi-Structured

The researcher uses guiding questions but allows flexibility.

Unstructured

The interview is flexible and develops through conversation.

Example: Semi-Structured Interview

Research Topic

Citizens' experiences with digital public services.

Possible interview questions:

- ▶ How do you usually access digital public services?
- ▶ What challenges have you experienced?
- ▶ What factors make the services easier or more difficult to use?
- ▶ How satisfied are you with the service?
- ▶ What improvements would you suggest?

Observation

Observation involves systematically watching and recording people, events, behaviors, or situations.

It may focus on:

- ▶ Human behavior
- ▶ Social interaction
- ▶ Organizational processes
- ▶ Physical environment
- ▶ Service delivery processes

Types of Observation

Observation may be classified as:

- ▶ Participant observation
- ▶ Non-participant observation
- ▶ Structured observation
- ▶ Unstructured observation

Example

Observing the process through which citizens receive services from a government office.

Focus Group Discussion

A **Focus Group Discussion (FGD)** is a guided discussion among a group of participants on a specific topic.

The researcher or moderator guides the discussion.

FGDs are useful for understanding:

- ▶ Shared experiences
- ▶ Group perceptions
- ▶ Differences of opinion
- ▶ Social interaction
- ▶ Community perspectives

Example: Focus Group Discussion

Research Topic

Challenges in accessing local government services.

Possible participants:

- ▶ Service users
- ▶ Community members
- ▶ Women
- ▶ Youth
- ▶ Elderly citizens

The researcher facilitates the discussion but does not dominate it.

Document Analysis

Document analysis involves systematically examining existing written, visual, or recorded materials.

Documents may include:

- ▶ Laws and policies
- ▶ Government reports
- ▶ Meeting minutes
- ▶ Annual reports
- ▶ Newspapers
- ▶ Institutional records
- ▶ Historical documents

Using Secondary Data

Secondary data may be obtained from:

- ▶ Census reports
- ▶ Government databases
- ▶ Administrative records
- ▶ Statistical reports
- ▶ International organizations
- ▶ Research institutions
- ▶ Previous studies
- ▶ Public databases

Important

Secondary data should be evaluated for relevance, quality, accuracy, and suitability for the current research.

Choosing an Appropriate Data Collection Method

The choice of method depends on:

- ▶ Research problem
- ▶ Research objectives
- ▶ Research questions
- ▶ Research design
- ▶ Type of data required
- ▶ Characteristics of participants
- ▶ Available resources
- ▶ Time available
- ▶ Ethical considerations

Matching Research Questions and Methods

Research Question	Possible Data Collection Method
How many citizens use a service?	Survey / Questionnaire
What factors influence satisfaction?	Survey / Structured Questionnaire
How do citizens experience the service?	Interview
How do people interact in a service environment?	Observation
What are community perceptions?	Focus Group Discussion
How has a policy developed over time?	Document Analysis

Pilot Testing

Before conducting the main study, researchers may conduct a **pilot test**.

A pilot test helps identify:

- ▶ Unclear questions
- ▶ Problems in instructions
- ▶ Difficulties in data collection
- ▶ Length of the instrument
- ▶ Problems with response options

Purpose

To improve the research instrument and data collection procedure before the main study.

Ensuring Data Quality

Good data collection should aim to ensure:

Validity

Does the instrument measure what it is intended to measure?

Reliability

Does the instrument produce consistent results?

Accuracy

Are the data correctly recorded?

Completeness

Is all necessary information collected?

Ethical Issues in Data Collection

Researchers should consider:

- ▶ Informed consent
- ▶ Voluntary participation
- ▶ Privacy
- ▶ Confidentiality
- ▶ Anonymity
- ▶ Protection from harm
- ▶ Honest representation of information
- ▶ Secure handling of data

Research Instruments

A **research instrument** is a tool used to collect data.

Examples include:

- ▶ Questionnaire
- ▶ Interview guide
- ▶ Observation checklist
- ▶ Focus group discussion guide
- ▶ Document review checklist

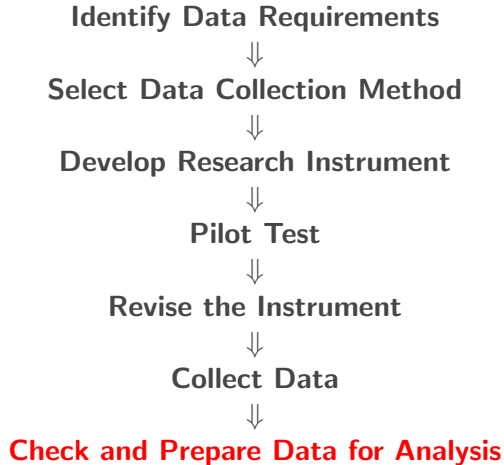
Important

The research instrument should be directly connected to the research questions, objectives, and variables or concepts being studied.

Example: Linking Objectives to Data Collection

Research Objective	Possible Method
To measure citizen satisfaction	Questionnaire / Survey
To identify factors influencing satisfaction	Survey + Statistical Analysis
To explore citizens' experiences	Semi-structured Interview
To understand group perceptions	Focus Group Discussion
To examine policy provisions	Document Analysis

The Data Collection Process



Key Takeaway

Good research depends on good data.

Research Question



Type of Data



Data Collection Method



Research Instrument



Data Analysis